

SEDAR 27: Gulf of Mexico Menhaden

Data Needs:

Collection of structures from gillnet surveys – Need to start collecting scales from gulf menhaden captured during the state gillnet surveys. Collection of scales would allow for the age of individuals to be determined in order to provide gillnet survey age composition data for the stock assessment.

Adult Monitoring Survey - Need to expand existing sampling protocols or develop additional protocols to monitor adult populations that specifically target adult menhaden inshore. Aerial surveys may be useful tools with ground-truthing for size and age.

Standardized Juvenile Index Sampling - Design and implement a survey dedicated to determining menhaden recruitment in the rivers and upper bays of the northern Gulf of Mexico.

Maturity and Fecundity - The seminal study on fecundity and sexual maturity of gulf menhaden was published thirty years ago (Lewis and Roithmayr 1981) with data from the late 1970s. It is recommended that a study should be initiated to re-examine the reproductive biology of gulf menhaden in the northern Gulf of Mexico, which includes updating fecundity estimates, maturity schedules, and sex ratios. Any study needs to reinvestigate whether gulf menhaden are determinate or indeterminate spawners.

Understanding Predator/Prey Relations - Expand the diet and stable isotope database to determine the trophic role of gulf menhaden in the northern Gulf of Mexico. Investigate fatty acids profiles as an additional more specific indicator of important prey items of gulf menhaden.

Most data available for *Brevoortia spp.* feeding behavior is based on examination of Atlantic menhaden (*B. tyrannus*). One key research need is that data on gulf menhaden feeding be collected to improve the specificity of ecosystem models. This includes direct analysis of diet, as well as examinations of feeding behavior, in response to key prey items. Direct diet enumeration is difficult due to the planktonic nature of the prey, but biochemical techniques such as analysis of stable isotope ratios (Litvin and Weinstein 2004, Rooker et al. 2006) and fatty acid profiles (Rooker et al. 1998), provide valuable tools for diet analysis of filter feeders. These techniques can also be used to examine the role of gulf menhaden as a

prey item for higher trophic level piscivores, which will allow for a more precise inclusion of menhaden in food web models of the Gulf of Mexico. An emphasis on quantifying the trophic role of menhaden in the Gulf of Mexico is an important step in the move towards ecosystem-based management.

Genetics - There is a need for further research on gulf menhaden stock structure, with an emphasis on increased genetic sampling (i.e. larger nuclear DNA marker data sets). More specifically, priority areas should include:

1. Identification in the Clupeid literature of potential new heterologous nuclear DNA markers (preferably microsatellites or SNP's) which will potentially enhance genetic sampling in gulf menhaden.
2. Identification of menhaden-specific nuclear DNA markers (preferably microsatellites or SNP's) using a lab-based DNA library screening technique.
3. Evaluation of the markers identified in (1) and (2) for appropriateness in population genetic studies of gulf menhaden.
4. Reassessment of gulf menhaden samples throughout the range of the species using a larger, more informative genetic panel of markers than that described in Anderson (2006).

Tagging Studies - Re-institute the gulf menhaden tag/recovery study. Many more tools exist today to simplify tag/recapture of fishes, and an updated tag/recapture study would allow for the estimation of natural mortality. Generally, natural mortality is one of the most difficult values in a stock assessment to determine, thus empirical evidence of the natural mortality rate would be beneficial. In addition, redoing the natural mortality study of Ahrenholz (1981) would provide information on whether or not the natural mortality rate is changing through time and whether it is increasing or decreasing.

Further Analyses and Modeling Approaches:

Fishery-independent data – Further evaluation of the available fishery-independent data and exploration of ways to combine the data from each state in order to provide a single coastwide index would benefit the stock assessment by providing information on trends in abundance over time.

Environmental factors – Exploration of environmental factors that play a crucial role in gulf menhaden recruitment dynamics and catchability (both fishery-dependent and fishery-independent) would be beneficial. Relationships related to recruitment could be applied to the stock-recruitment curve in the model to better define the number of recruits produced each year. The effects that environmental factors have on catchability of different fishery-independent and fishery-dependent gears would provide

information to the model on if catchability is changing over time and how, which will lead to better estimates of abundance and trajectory over time.

Establish additional research of simulation models to incorporate the fishery into ecological scenarios which may include MSVPAs, ECO-SIM, EcoPath, etc. to get better estimates of natural mortality, which would account for predator-prey dynamics.

Review Workshop

Although results were unsatisfactory for this stock assessment, they did serve to clarify additional research necessary for future assessment efforts. Prioritized lists of short- and long-term research recommendations are presented below.

Prioritized list of short-term research recommendations:

Adult abundance index: Review methods that could be used to provide a reliable fishery-independent adult abundance time series. A pilot survey should be implemented as soon as possible. Development of a long-term time series is needed to increase the certainty of menhaden stock assessments.

Analysis of CDFR data: These data may contain an abundance signal on a weekly and/or an annual basis. In the long-term, the data should be fully analyzed in this regard. In the short-term, a standardized CPUE time series should be developed from the data for use in stock assessment.

Further analysis of fishery-independent state indices: These data need to be fully analyzed with regard to determining the best methods to use the data to provide potential juvenile and adult abundance indices.

Ageing: The consistency of the age readings throughout the whole time series should be checked. The current reader has read scales since 1969 and there may be some drift in her readings. Also, other readers participated up to the early 1970s and there is evidence of relative bias in the readings up to 1970 which should be investigated.

Further development of the SRA: The incorporation of catch-at-age data into the SRA approach is encouraged as this would allow the method to provide a stand-alone stock assessment for menhaden.

Prioritized list of long-term research recommendations:

Adult abundance survey: The existing state sampling of coastal waters is not adequate for providing a defensible adult abundance index. In the absence of such an index, stock assessment of menhaden will

continue to be problematic. The development of a fishery-independent adult-abundance index should be given a very high priority. A review of possible methods is the first step (see short-term research recommendations above). Aerial surveying using visual estimation and/or LIDAR should be considered among the options.

Biological data: All biological parameters pertinent to the stock assessment should be updated. Subsequently, they should be monitored every few years.

Catch sampling: The potential bias associated with sampling only the last catch of the day should be investigated. It is important to know if there could be a bias and whether it is towards larger/older fish or smaller/younger fish.